

# MOHSEN MOSTAFA

Independent Researcher — Physical & Geometric AI | AI Team Lead & Research Engineer

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## RESEARCH STATEMENT

I define and develop Physical & Geometric AI — a unified methodology for building robust, trustworthy deep learning systems under computational constraints. My research integrates three pillars: (1) geometric foundations (proving how networks adapt to data manifolds), (2) uncertainty-aware architectures (Bayesian normalization and activation functions), and (3) physics-informed optimization (curvature-aware optimizers for scientific ML). I completed six first-author papers forming this methodology in under 10 weeks using only free cloud GPUs (Kaggle/Colab: NVIDIA T4 15GB, P100 16GB), demonstrating that high-impact research is achievable without massive computational resources. I am currently extending this methodology toward embodied prediction, including geometric priors for RGB-D representation learning and safe routing for adaptive world models.

## EDUCATION

**B.Sc. Management Information Systems** — High Institute for Specialized Technological Studies, Cairo, Egypt (2000 – 2005)

## RESEARCH EXPERIENCE

**Independent Researcher — Physical & Geometric AI** | *Self-directed*

2023 – Present

- Founded and executed a coherent research program resulting in six first-author publications within 10 weeks, all validated on free cloud GPUs (Kaggle/Colab).
- Defined the “Physical & Geometric AI” methodology, unifying geometric deep learning theory, uncertainty-aware architectural components, and physics-informed optimization.
- Proved four fundamental theorems on how gradient descent shapes neural representations on data manifolds.
- Developed novel Bayesian normalization layers (R-LayerNorm, Bayesian R-LayerNorm) and activation functions (Bayesian PASA) with built-in uncertainty quantification.
- Created EPANG-Gen, a memory-efficient curvature-aware optimizer that eliminates catastrophic training failures on stiff PDE problems.
- Validated all methods on challenging benchmarks: turbulent flow, CIFAR-10-C, and Darcy flow — under strict computational constraints.
- Built SafePredict, a real-time “shadow monitor” for humanoid robot safety bridging open-source robotics data (LeRobot v0.6.0) with enterprise MLOps (Triton/Kafka) to catch physical-AI failures before they reach actuation. [Project](#)
- Currently scoping two new papers extending this methodology toward embodied prediction: LePhyJEPA (geometric priors in latent predictive architectures for RGB-D representation learning) and Graph-Grounded Safe Routing for Adaptive World Models, the first paper in a broader research program on cognitive architectures for video understanding (SARI).

## PUBLICATIONS & PREPRINTS

### *Geometric Foundations*

- Mostafa, M. — Geometric Foundations of Racing Dynamics: How Gradient Descent Adapts Network Capacity on Data Manifolds with Application to Bayesian R-LayerNorm. (2026). Q1 Journal (Preprint).
- Mostafa, M. — Gaussian-Constrained LeJEPA Representations for Unsupervised Scene Discovery in the Image Matching Challenge 2025. (2025). [arXiv:2602.07016](https://arxiv.org/abs/2602.07016).
- Mostafa, M. — HPST: A Hybrid Physics-Spectral-Threshold Framework for Fluid Flow Analysis with Theorem Proving and Graph Neural Networks. (2026). Preprint.

### *Uncertainty-Aware Architectures*

- Mostafa, M. — Bayesian R-LayerNorm: Uncertainty-Aware Adaptive Normalization with Provable Robustness Bounds. (2026).
- Mostafa, M. — Bayesian PASA: Probabilistic Adaptive Sigmoidal Activation with Uncertainty Quantification. (2026).
- Mostafa, M. — Uncertainty-Aware Classifier with Physics-Based Rejection (UA-PBR): A Proof-of-Concept under Computational Constraints. (2026).
- Mostafa, M. — R-LayerNorm: Robust Layer Normalization with Adaptive Noise Suppression for Noisy Image Data. (2025). Preprint.

### *Physics-Aware Optimization*

- Mostafa, M. — EPANG-Gen: A Curvature-Aware Optimizer with Uncertainty Quantification for Scientific Machine Learning — A Proof-of-Concept under Computational Constraints. (2026).

### Under Review

- Mostafa, M. — Geometric Deep Learning for Point Cloud Segmentation. Under revision for WACV 2027.
- Mostafa, M. — Physics-Informed Neural Networks for Fluid Flow. (2025). arXiv.

## PROFESSIONAL EXPERIENCE

### AI Team Lead & Computer Vision Researcher | GraviLog (Remote — US/UAE) Sep 2025 – Present

- Designed end-to-end multimodal AI system architecture (CV + LLM) with TensorRT/ONNX optimizations achieving 40% latency reduction.
- Defined technical roadmap for 3 products and delivered strategic presentations to C-level stakeholders.
- Led RAG retrieval pipeline (Pinecone + embeddings); mentored 4 engineers in MLOps and system design.

### Senior AI Developer (Research Focus) | OMC Technologies (Remote — Limerick, Ireland) May 2023 – Aug 2025

- Built AI agents integrating LLMs (Falcon, DeepSeek, Qwen, Llama) — automated 3 workflows, reducing manual effort by 40%.
- Deployed production RAG system (LangChain + Pinecone) reducing hallucinations by 35%.
- Multimodal healthcare AI system: F1 score +19%; latency reduced from 210ms to 145ms via streaming + FP16.
- Fraud detection system: AUC +17%, false positives -22%; FastAPI service with  $p99 \leq 95\text{ms}$  at 500 RPS.
- Managed AI infrastructure with 24 GPUs across 12 nodes for real-time IoT anomaly detection.
- Mentored 5 junior engineers, leading to 2 promotions.

### AI Developer & Researcher | Nexgro (South Africa) Apr 2020 – Apr 2023

- Grain quality CV pipeline achieving 98% accuracy across 6 grain types.
- Drone/satellite anomaly detection (NDVI + YOLOv8): manual inspection efficiency +160%.
- Optimized 3D point cloud segmentation for LiDAR/DSM data achieving sub-300ms inference on edge devices.
- Implemented TensorRT/ONNX optimizations delivering 3× speedup in model inference.
- LLM RAG assistant on technical manuals — saved 15 person-hours/week.

### Data Scientist | Nexgro (South Africa) Mar 2018 – Mar 2020

- Predictive ML models for land use classification, crop yield estimation, and water consumption forecasting.
- Forecast models reduced logistics costs by 18%; predictive maintenance prevented 3 breakdowns (~\$45K saved).
- Engineered ETL pipelines reducing data processing time by 20%.

### Systems Administrator | Egyptian Co. Jun 2013 – Mar 2018

- IT infrastructure management for 200+ users → 99.8% uptime; multi-site network & servers.
- Automation scripts reduced manual ticket handling by 25%.
- Self-directed pivot to data science (certifications 2016–2018); managed LAN/WAN, firewalls, backup for 150+ nodes.

### IT Network Administrator | International Co. for Trade Aug 2009 – Jun 2013

- Administered LAN/WAN, firewalls, and backup solutions for 150+ nodes.

### IT Network Specialist | International Co. for Trade Aug 2005 – Jun 2009

- First-line network support, hardware procurement, and vendor coordination.

## TECHNICAL SKILLS

- Deep Learning: PyTorch, TensorFlow, Hugging Face (Falcon, Llama, Mistral, Qwen), JEPA, CLIP, YOLO, Mask2Former
- Probabilistic ML: Bayesian Deep Learning, Uncertainty Quantification, Natural Gradient Methods, Curvature-Aware Optimization
- GenAI / LLM: RAG, ColBERT, LangChain, DeepSeek, RLHF concepts, Streaming Inference
- Computer Vision: OpenCV, 3D Point Cloud Segmentation (LiDAR/DSM), Multi-View Geometry, Image Processing
- MLOps & DevOps: MLflow, Kubeflow, Airflow, Docker, Kubernetes, KServe, FastAPI, GitLab CI, GCP Vertex AI, AWS SageMaker
- Streaming / Big Data: Apache Kafka, Spark Structured Streaming, Hadoop HDFS, Parquet
- Optimization: TensorRT, ONNX, Model Pruning/Quantization, FP16 Inference
- Data & Databases: Python (Pandas, Dask, Polars, PySpark), SQL (PostgreSQL), MongoDB, Redis, Pinecone
- Responsible AI: GDPR, AI Ethics, Fairness, Transparency, Uncertainty-Aware Systems

## SELECTED CERTIFICATIONS

- Deep Learning Specialization — DeepLearning.AI
- TensorFlow Developer & Advanced TensorFlow — Google
- Google Cloud MLOps — Google
- Total Data Quality — University of Michigan
- Research Methodology — University of London
- IBM Data Science Professional Certificate — IBM
- CCNA 200-301 & ENCORE 350-401 — Cisco

## LEADERSHIP & COMMUNITY

- Mentored 5+ engineers across OMC Technologies and GraviLog; 2 direct promotions.
- Conducted internal LLM/MLOps workshops and knowledge-transfer sessions.
- Designed full MLOps showcase pipeline: Kafka → Spark Streaming → HDFS → MongoDB → PostgreSQL → MLflow → Kubeflow → FastAPI for real-time predictive maintenance.
- Advocate for resource-constrained AI research; demonstrated 6-paper research program achievable on free-tier cloud GPUs.

## LANGUAGES

- Arabic — Native
- English — Fluent (C1/C2)